Getting started with Python

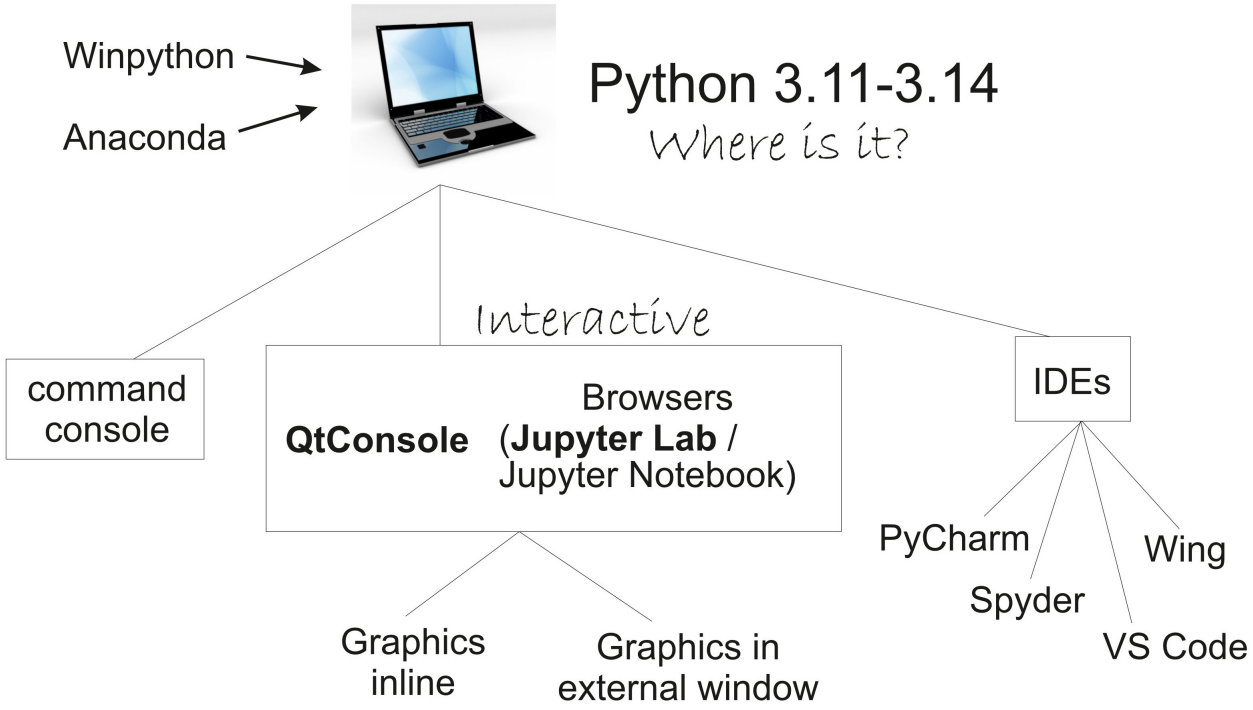
Quick hints

Thomas Haslwanter

Department of Medical Engineering
University of Applied Sciences Upper Austria

Master Medical Technology, 2026

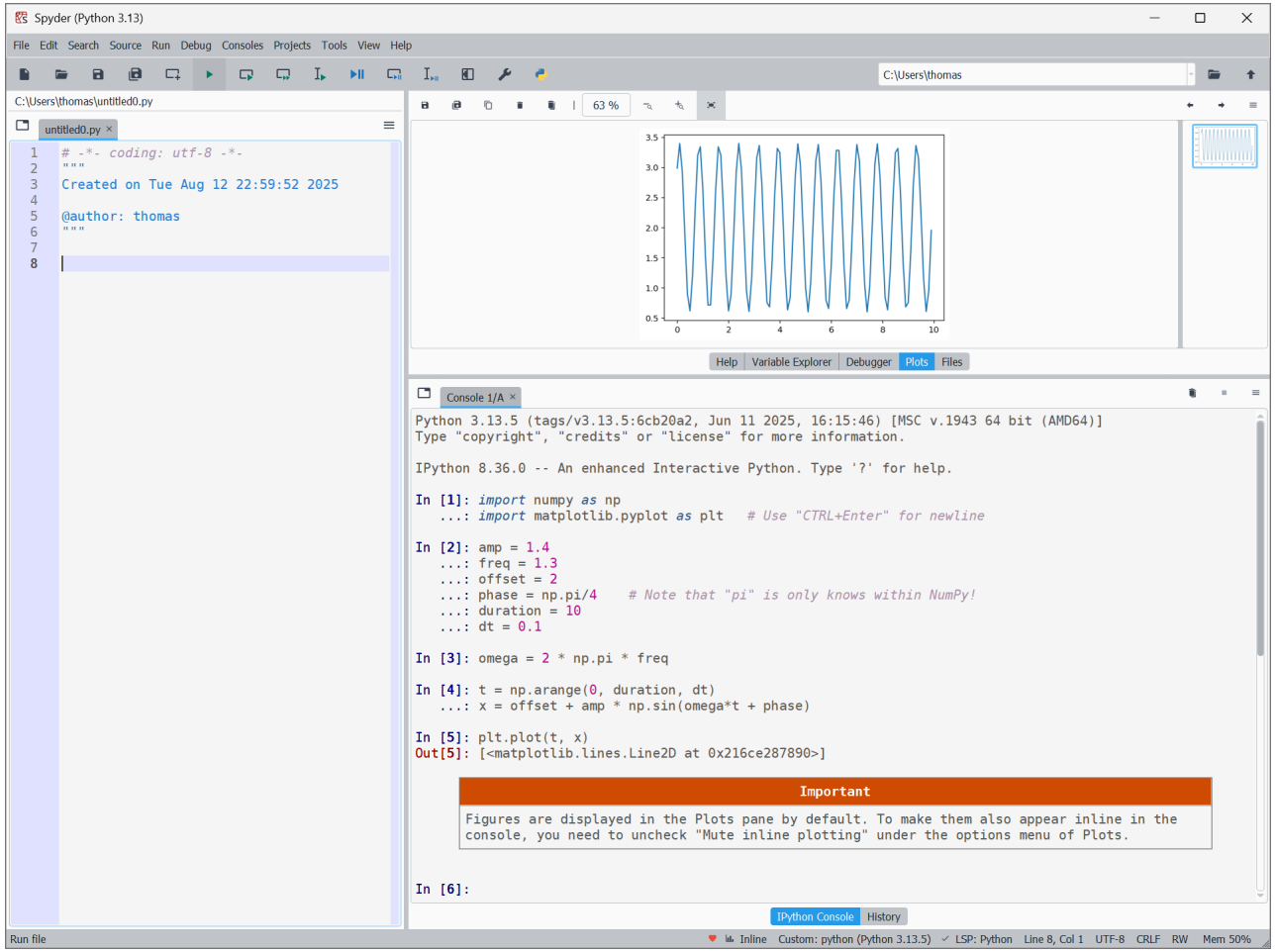
Installation



Get it from http://winpython.github.io/ or from https://anaconda.org/.

For Windows users *WinPython* is recommend it. You best use *Python 3.14.5*, and install it into a separate directory. I suggest that you use (or create) the directory: `C:/Programs`

Plotting a sine-wave



Start `Spyder` and try to implement the example as shown above. Make sure that you are using Spyder version 6.x .

Python and Scientific Packages.

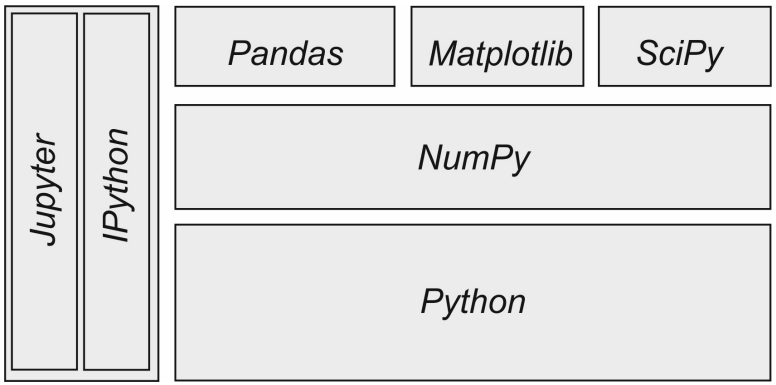
Numpy, Scipy, Matplotlib.

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Python and *packages*



Scientific Python Packages

Numpy	numerical algorithms
Matplotlib	plotting
Scipy	scientific computations (FFT, integration, ...)
Pandas	data structures
IPython	A powerful *interactive* shell for Python
Jupyter	Open-source applications for interactive and collaborative programming

Python and Scientific Packages.

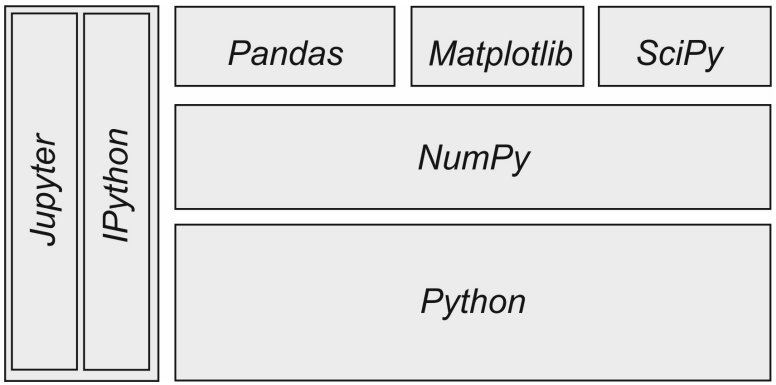
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"Python in Seven Steps"

https://work.thaslwanter.at/py_intro/



I have prepared a gradual introduction to Python, in seven managable steps. It contains instructions that *YOU* should follow, and links to helpful resources, and videos to support you when you get stuck.

Help: If you have no clue where to start ...

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1. Search only in:
   - http://stackoverflow.com/
   - https://www.python.org/doc/
   - http://docs.scipy.org/
   - https://matplotlib.org

7 Jupyter/IPython Tips

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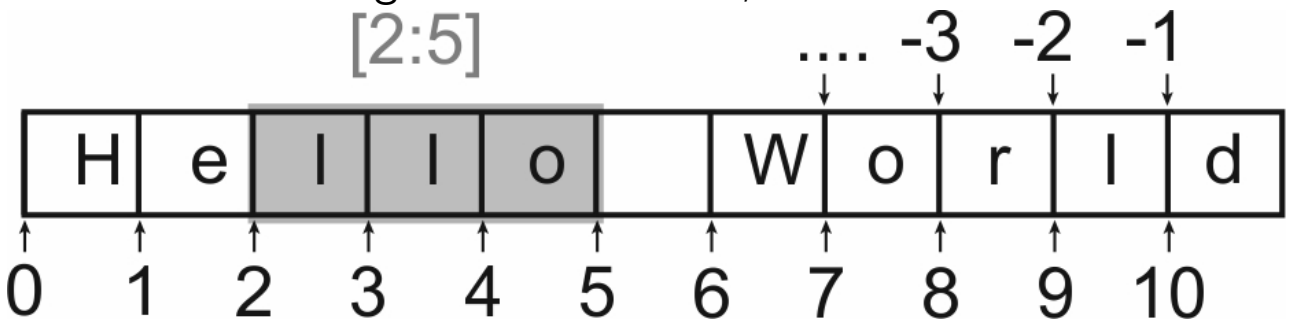
1. Start with an interactive interface (some implementation of IPython, e.g. `jupyter [qtconsole/notebook/lab]` or `Spyder`).
2. For help on e.g. `plot`, use `plot?` or `help(plot)`.
3. Check out help tips with `%quickref`.
4. Customize IPython on your computer, and create batch-files to. start `jupyter lab` and `jupyter qtconsole`. This will take quite a bit of care and effort - but is well worth while, and will save you lots of time in the long run! (Check my book Hands-on Signal Analysis with Python for more infos.)
5. TAB-completion, for file- and directory names, variable names, AND for commands.
6. To switch between inline and external graphs, use `%matplotlib inline` and `%matplotlib`.
7. Use `%runfile [aFile]` to run a file, and `import [aFile]` to import it.

7 Python Tips

1. Stick to the standard conventions (- and use templates!).
2. Everything in Python is an object: to find out about "obj", use `type(obj)` and `dir(obj)`.
3. Learn to use the debugger!
4. Know *lists*, *tuples*, and *dictionaries*, as well as *panda's DataFrames*.
5. *pip list* to check what is installed, *pip show [package-name]* to see package details, and *pip install [xxx] [-U]* to install/update packages.
6. Use functions a lot, and understand the `if __name__=='__main__':` construct.
7. To get the current directory, use `os.path.abspath(os.curdir)`.
   In IPython: `%cd` or `cd`

7 NumPy Tips

1. Use arrays, not matrices.
2. Python is vector oriented. For example, `np.ones(3)` gives you a vector! `np.ones((3,3))` gives you a 3x3 matrix of ones. (This is different from Matlab, which is matrix oriented.)
3. Create row- and column vectors with `np.r_[data]` and `np.c_[[data]]`
4. A 2-array with one row is different from a vector!
5. Learn about broadcasting (e.g. `my_mat + my_vector`)
6. Transpose a matrix with ".T"
7. Learn about "slicing": it starts with 0, and the last element is not included!



7 Matplotlib Tips

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1. In Python scripts, you have to use `plt.show()` to display the current figure! (Spyder is by default in `interactive` mode, so you don't need it there. But put it in anyway: you need it when scripts are run standalone!)
2. Get used to the object oriented Matplotlib-style: it is more efficient and clearer.
3. Label lines right away.
4. Learn how to switch between inline and external graphs.
5. Use the Matplotlib gallery for code-tips: http://matplotlib.org/gallery.html
6. Use `axhline` and `axvline` for horizontal/vertical lines.
7. Use `fig, axs = plt.subplots(3,2)` for subplots.

Links

Installation-Windows https://winpython.github.io (Install it in C:/Programs.)

Installation-Win/Mac/Linux https://anaconda.org

Quickstart- 4 hours https://github.com/hrzn/4hours-python-intro

Quickstart- 7 steps https://work.thaslwanter.at/py_intro

Scipy Lecture Notes https://scipy-lectures.org

Beginner Tips https://realpython.com/python-beginner-tips

NumPy for Matlab Users https://docs.scipy.org/doc/numpy/user/numpy-for-matlab-users.html

SaPy github repo https://github.com/thomas-haslwanter/sapy

Learn Python https://www.learnpython.org/

Books

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General Python Book Python Crash-Course (Matthes, 3rd Ed, 2023): The most frequently recommended book. General Python, no numerical analysis.

Numerical Computation Numerical Python (Johannson, 3rd Ed, 2024): Very complete, contains more than we need here. But a very good overall book on signal analysis. It starts from scratch, and includes quite advanced topics such as machine learning.

Exercise 1: 'Hello, World'

- Perform "Hello, World" in *IPython*.
- Make a module called "hello.py", and run it from the command-line.
- Run this module in an IDE, e.g. Spyder .
- Run the module in *Jupyter*.

Tips:

- Use `%run hello.py` or `run hello` to run this module in the *IPython*.
- Add `input('Done')` to hold the module before it closes the console.

Exercise 2: *numpy*

- Write a program that displays the numbers from 0 to 90, in step of 10, and the sine of each of these numbers (in degrees).
- E.g.:
  "The sine of 0 deg is 0."
  "The sine of 10 deg is 0.17."
  ...

Tips:

- `import numpy as np`
- `np.arange(0,6,2)` -> 0, 2, 4 "up to, but not including the last value!"
- `print(f'pi is {np.pi:.2f}')`
- `for ii in ... :` loops are defined by the indentation!

Exercise 3: Functions

Turn the code from *Exercise 2* into a function, where:

- The upper limit is the input argument (default = 90).
- The number of lines written is the return argument.

Tips:

- `def sineValues(upperLim=90):`
- functions are defined by indentation
- `return ....`

Exercise 4: *matplotlib.pyplot*

- First plot any sine wave.
- Then plot 3 seconds of a noisy sine-wave with 2 Hz, an amplitude of 5, a sampling rate of 100 Hz, and a phase shift of 30 deg.
- Overlay this with a clean sine with the same parameters, but in a thick, red line.
- For help, check out the sample gallery of Matplotlib.

Tips:

- `import matplotlib.pyplot as plt`
- Vector generation: `x = np.arange(5,10,0.1)`
- Length of object: `len(t)`
- Use "SHIFT+Enter" for entering multiple commands (e.g. plot-superposition) in *IPython*.